

Curriculum Vitae

Full Name: David Andrew Snyder

Current Position: Postdoctoral Research Fellow
Electrical and Systems Engineering
University of Pennsylvania

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EDUCATION

Princeton University	2019-2025
Ph.D. – Mechanical and Aerospace Engineering	
M.A. – Mechanical and Aerospace Engineering	
University of Maryland, College Park	2014-2018
B.S. – Aerospace Engineering	
B.S. – Economics	

COLLABORATIONS AND FUNDING

Toyota Research Institute (TRI)	AY 2025
Graduate Research Intern	
Toyota Research Institute (TRI)	Summer 2024
Google Artificial Intelligence Lab	2021-2022

HONORS AND AWARDS

Penn AI Fellowship	Jan 2026 - Present
National Science Foundation Graduate Research Fellowship	2019-2024
Sir Gordon Y. S. Wu Fellowship in Engineering	2019-2024
Crocco Award for Excellence in Teaching	Fall 2022

PROFESSIONAL SERVICE

Reviewer	
International Conference on Robotics and Automation (ICRA)	(2022 - Present)
Robotics and Automation Letters (RA-L)	(2023 - Present)
International Journal of Robotics Research	(2023)

Workshop Organizer
(RSS 2025) *2nd Workshop OOD Generalization in Robotics*
(CoRL 2023) *OOD Generalization in Robotics: Towards Reliable Learning-Based Autonomy*

TEACHING ASSISTANTSHIPS

[4] (Upper-Level) Introduction to Robotics	Fall 2021
[3] Introduction to Engineering Dynamics	Spring 2021, 2022
[2] (Upper-Level) Control of Aerospace Systems	Spring 2018
[1] First-year Independent Research Experience: Autonomous Unmanned Systems	Spring 2016

PUBLICATIONS AND PRESENTATIONS

Guide: *equal author contribution; †under review; **bold text** denotes D. Snyder presented.

Journal Papers

- [†] Lihan Zha, Apurva Badithela, Michael Zhang, Justin Lidard, Jeremy Bao, Emily Zhou, David Snyder, Allen Z. Ren, Dhruv Shah, and Anirudha Majumdar. “Guiding Data Collection via Factored Scaling Curves;” *Under Review*.
- [2] Nathaniel Simon, Alexander Pique, David Snyder, Kyle Ikuma, Anirudha Majumdar, and Marcus Hultmark. “Fast-Response Hot-Wire Flow Sensors for Wind and Gust Estimation on UAVs”, *Measurement Science and Technology* 34 (2023) 025109.
- [1] Justin M. Lidard, Debdipta Goswami, David Snyder, Girguis Sedky, Anya R. Jones, and Derek A. Paley. “Feedback Control and Parameter Estimation for Lift Maximization of a Pitching Airfoil”, *Journal of Guidance, Control, and Dynamics* (2021) 44:3, 587-594.

Conference Papers

- [†] Zachary Ravichandran, David Snyder, Alexander Robey, Hamed Hassani, Vijay Kumar, and George J. Pappas. “Contextual Safety Reasoning and Grounding for Open-World Robots;” *Under Review*.
- [†] David Snyder, Apurva Badithela, et. al. “Beyond Binary Success: Sample-Efficient and Statistically Rigorous Robot Policy Comparison;” *Under Review*.
- [†] Apurva Badithela, David Snyder, Lihan Zha, Joseph Mikhail, Matthew O’Kelly, Anushri Dixit, and Anirudha Majumdar. “Reliable and Scalable Robot Policy Evaluation with Imperfect Simulators;” *Under Review*.
- [8] **David Snyder**, Asher James Hancock, Apurva Badithela, Emma Dixon, Patrick Miller, Rares Andrei Ambrus, Anirudha Majumdar, Masha Itkina, and Haruki Nishimura. “Is Your Imitation Learning Policy Better than Mine? Policy Comparison with Near-Optimal Stopping;” *Proceedings of Robotics: Science and Systems (RSS)* (2025).
- [7] Eric Lepowsky*, **David Snyder***, Alexander Glaser, and Anirudha Majumdar. “Privacy-Preserving Map-Free Exploration for Confirming the Absence of a Radioactive Source”; *IEEE International Conference on Intelligent Robots and Systems* (2024).
- [6] **David Snyder**, Meghan Booker, Nathaniel Simon, Wenhan Xia, Daniel Suo, Elad Hazan, and Anirudha Majumdar. “Online Learning for Obstacle Avoidance”; *7th Annual Conference on Robot Learning (CoRL)* (2023).
- [5] Nathaniel Simon, Allen Z. Ren, Alexander Piquè, David Snyder, Daphne Barretto, Marcus Hultmark, and Anirudha Majumdar. “FlowDrone: Wind Estimation and Gust Rejection on UAVs Using Fast-Response Hot-Wire Flow Sensors”; *Proceedings of IEEE International Conference on Robotics and Automation (ICRA)* (2023).
- [4] Alec Farid*, **David Snyder***, Allen Z. Ren, and Anirudha Majumdar. “Failure Prediction with Statistical Guarantees for Vision-Based Robot Control”; *Proceedings of Robotics: Science and Systems* (2022).
- [3] Udaya Ghai*, **David Snyder***, Anirudha Majumdar, and Elad Hazan. “Generating Adversarial Disturbances for Controller Verification”; *Proceedings of the 3rd Conference on Learning for Dynamics and Control* (2021), PMLR 144:1192-1204.
- [2] Justin Lidard, Debdipta Goswami, David Snyder, Girguis Sedky, Anya Jones, and Derek A. Paley. “Output Feedback Control for Lift Maximization of a Pitching Airfoil”, *AIAA Guidance, Navigation, and Control Conference, AIAA SciTech Forum*, (AIAA 2020-1836).
- [1] Nathan Lauer, Derrick W. Yeo, David A. Snyder, and Derek A. Paley. “Tip-Vortex Localization for Cross-Stream Position Control of a Multi-Hole Probe Relative to a Stationary Wing in a Free-Jet Wind Tunnel”, *AIAA Guidance, Navigation, and Control Conference, AIAA SciTech Forum*, (AIAA 2017-1484).

Workshop Submissions

- [3] Apurva Badithela, David Snyder, Lihan Zha, Joseph Mikhail, Matthew O’Kelly, Anushri Dixit, and Anirudha Majumdar. “Reliable and Scalable Robot Policy Evaluation with Imperfect Simulators;” *CoRL*

2025: *Evaluation and Deployment Across the Robot Learning Lifecycle Workshop*; September 2025.

Winner, Best Paper Award.

[2] Eric Lepowsky* and David Snyder*. “Realizable Deployment of Limited-Knowledge Robotic Inspectors for Nuclear Verification,” *ICRA 2023: Workshop on Bridging the Lab-to-Real Gap*; June 2023.

[1] Nathaniel Simon, Allen Z. Ren, Alexander Pique, David Snyder, Daphne Barretto, Marcus Hultmark, and Anirudha Majumdar. “FlowDrone: Wind Estimation and Gust Rejection on UAVs Using Fast-Response Hot-Wire Flow Sensors,” *CoRL 2022: Learning for Agile Robotics Workshop*; December 2022.